

Multiple Choice

1. **Answer:** A

Options: A) $(p^2) / (mRT)$; B) $(RT) / (pV)$; C) $(nRT) / (pV^2)$; D) $(mRT) / p^2$; E) $(RT) / p^2$

Note: Correct option: A - $(p^2) / (mRT)$

2. **Answer:** D

Options: A) $f(x) \sim (1/2)c + (2c/\pi^2)\infty \sum_k = 1[\{cos[(2k)/c]\}/(2k)^2]$; B) $f(x) \sim (1/2)c + (4c/\pi^2)\infty \sum_k = 0[\{cos[(2k+1)/c]\}/(2k+1)^2]$; C) $f(x) \sim (1/2)c - (4c/\pi^2)\infty \sum_n = 1[1/n^2]cos\{(n)/c\}$; D) $f(x) \sim (1/2)c - (4c/\pi^2)\infty \sum_k = 0[\{cos[(2k+1)/c]\}/(2k+1)^2]$; E) $f(x) \sim (1/2)c - (4c/\pi^2)\infty \sum_k = 1[\{cos[(2k)/c]\}/(2k)^2]$

Note: Correct option: D - $f(x) \sim (1/2)c - (4c/\pi^2)\infty \sum_k = 0[\{cos[(2k+1)/c]\}/(2k+1)^2]$

3. **Answer:** B

Options: A) $2\delta(t) + (3e^{-t} + e^{-3t})u(t)$; B) $2\delta(t) + (3e^{-t} + 2e^{-3t})u(t)$; C) $3\delta(t) + (3e^{-t} + 2e^{-3t})u(t)$; D) $2 + (3e^{-t} + 2e^{-3t})u(t)$; E) $2 + (5e^{-t} + 2e^{-3t})$

Note: Correct option: B - $2\delta(t) + (3e^{-t} + 2e^{-3t})u(t)$

3. **Answer:** D

Options: A) 1.27 ohm; B) 1.07 ohm; C) 0.107 ohm; D) 0.77 ohm; E) 0.87 ohm

Note: Correct option: D - 0.77 ohm

4. **Answer:** C

Options: A) 6; B) 1; C) 2; D) 5; E) 7

Note: Correct option: C - 2

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '250 Hz'.

8. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

9. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.
10. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** .000037. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.
12. **Answer:** ($S^2 / 12$). Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key